

MECHANICS

Momentum / Collisions

$$p = mv$$

$$E_k = \frac{1}{2}mv^2$$

$$E_k = \frac{p^2}{2m}$$

$$J = Ft = \Delta p$$

Circular Motion

$$\theta = \frac{s}{r}$$

$$\omega = \frac{2\pi}{T} = 2\pi f$$

$$v = r\omega$$

$$a = \frac{v^2}{r} = r\omega^2$$

$$F = \frac{mv^2}{r} = mr\omega^2$$

ELECTRIC FIELDS

$$E = \frac{F}{Q}$$

$$E = \frac{V}{d}$$

$$W = QV$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

$$E = -\frac{dV}{dr}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2}$$

Constants

$$\epsilon_0 = 8.85 \times 10^{-12}, \text{F m}^{-1}$$

$$k = 8.99 \times 10^9, \text{N m}^2\text{C}^{-2}$$

$$e = 1.60 \times 10^{-19}, \text{C}$$

CAPACITORS

$$C = \frac{Q}{V}$$

$$E = \frac{1}{2}QV$$

$$E = \frac{1}{2}CV^2$$

$$E = \frac{Q^2}{2C}$$

$$\tau = RC$$

$$Q = Q_0 e^{-t/RC}$$

$$V = V_0 e^{-t/RC}$$

$$I = I_0 e^{-t/RC}$$

MAGNETISM / ELECTROMAGNETISM

$$\Phi = BA \cos \theta$$

$$F = BIL \sin \theta$$

$$F = Bqv \sin \theta$$

$$r = \frac{mv}{Bq}$$

$$\frac{1}{2}mv^2 = qV$$

$$\varepsilon = -\frac{d(N\Phi)}{dt}$$

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

PARTICLE PHYSICS

$$\lambda = \frac{h}{p}$$

$$E = hf$$

$$E = mc^2$$

Constants

$$h = 6.63 \times 10^{-34}, \text{J s}$$

$$c = 3.00 \times 10^8, \text{m s}^{-1}$$

THERMAL PHYSICS

$$Q = mc\Delta T$$

$$Q = mL$$

$$E = \frac{3}{2}kT$$

$$pV = nRT$$

$$pV = NkT$$

Constants

$$R = 8.31, \text{ J mol}^{-1}\text{K}^{-1}$$

$$k = 1.38 \times 10^{-23}, \text{ J K}^{-1}$$

$$N_A = 6.02 \times 10^{23}, \text{ mol}^{-1}$$

NUCLEAR DECAY

$$A = \lambda N$$

$$N = N_0 e^{-\lambda t}$$

$$A = A_0 e^{-\lambda t}$$

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

OSCILLATIONS

$$f = \frac{1}{T}$$

$$F = -kx$$

$$a = -\omega^2 x$$

$$\omega = 2\pi f$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$x = A \cos(\omega t)$$

$$v = -A\omega \sin(\omega t)$$

$$a = -A\omega^2 \cos(\omega t)$$

GRAVITATION / ASTROPHYSICS

$$F = \frac{GMm}{r^2}$$

$$g = \frac{GM}{r^2}$$

$$V = -\frac{GM}{r}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$T = 2\pi\sqrt{\frac{r^3}{GM}}$$

Constants

$$G = 6.67 \times 10^{-11}, \text{N m}^2\text{kg}^{-2}$$

$$g = 9.81, \text{m s}^{-2}$$

STELLAR PHYSICS

$$L = 4\pi R^2 \sigma T^4$$

$$\lambda_{\max} T = b$$

Constants

$$\sigma = 5.67 \times 10^{-8}, \text{W m}^{-2}\text{K}^{-4}$$

$$b = 2.90 \times 10^{-3}, \text{m K}$$

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